

Erdős Problem #817

EP-817

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and define $g_k(n)$ to be the minimal N such that $\{1, \dots, N\}$ contains some A of size $|A| = n$ such that $\langle A \rangle = \{\sum_{a \in A} \epsilon_a a : \epsilon_a \in \{0, 1\}\}$ contains no non-trivial k -term arithmetic progression. Estimate $g_k(n)$. In particular, is it true that $g_3(n) \gg 3^n$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>